

A hydrogel modified electrode for application as a voltammetric temperature sensor and its use in oxygen detection

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We report a hydrogel modified electrode based voltammetric “thermometer” and its application in amperometric oxygen detection so avoiding the need for external temperature monitoring to allow the rigorous measurement of oxygen concentrations. A platinum electrode was modified with a layer of agarose hydrogel containing decamethylferrocene (DmFc) and bisferrocene (BisFc) and a layer of pure ionic liquid containing no ferrocenes as solvent. The differences of the formal potentials between these redox couples were measured using square wave voltammetry and a linear increase of peak separations with temperature was observed so providing a basis for measuring temperature. This system was then applied to the analysis of dry oxygen. The temperature was monitored either by a conventional external thermometer or by the internal voltammetry confined to the agarose hydrogel layer on the electrode surface. The concentration and diffusion coefficient of oxygen were obtained with excellent precision using chronoamperometry over a range of temperatures. Excellent agreement between the two approaches was found showing that the voltammetric redox couples provide a good basis for an *in situ* “thermometer”.

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1 Introduction

Electrochemistry offers lower cost, higher sensitivity and faster responses as compared to many analytical techniques.^{1,2} Electrochemical gas sensors are particularly attractive since they combine low cost with modest equipment demands where only a compact analytical cell and small electrical power supplier are required.³ Examples include detections of O₂,⁴ H₂,⁵ CO₂,⁶ NO₂ (ref. 7) and NH₃.⁸ Most commercially available gas sensors are amperometric based, where the level of gas present is indicated by the magnitude of current passed during the electroanalysis.⁹ The thermodynamics, kinetics and transport of such sensing systems are all highly influenced by the temperature which results in the need of frequent calibration of sensors against temperature. Therefore to overcome this, it is desirable to incorporate some sort of “thermometer” into these sensing systems. Various types of external thermometers have been developed.¹⁰ However instead of monitoring temperature directly at the electrode surface, such external probes are remote from the sensing system such that less accurate and slower measurement may result.³ Previously, work by Xiong *et al.*^{11,12} has developed proof of concept of an *in situ* voltammetric thermometer and has incorporated this thermometer into an oxygen gas sensor as a model. Ferrocenes were employed as internal temperature detectors and as internal

redox couples to avoid the need for elaborate reference electrodes, instead allowing potential control *via* a pseudo reference electrode. However, the ferrocene derivatives used were prone to some evaporative loss which potentially shortens the lifetime of the sensors.¹³ To minimise the loss of ferrocenes, the present work seeks to ‘immobilise’ ferrocene derivatives onto or near the electrode surface using an agarose hydrogel layer on the electrode.

Agarose hydrogel is cheap, non-toxic and simply applied. It has been successfully and widely used to entrap biological molecules (heme proteins,¹⁴ cytochrome c,¹⁵ horseradish peroxidase,¹⁶ ferritin,¹⁷ *etc.*) onto electrode surfaces. Following the success of previous work, this work continues to use room temperature ionic liquids as solvents for the basis of gas sensing. Room temperature ionic liquids (RTILs) are entirely composed of ions and exist as liquids at or below room temperature. Their exceptional properties, including high chemical and thermal stability, intrinsic conductivity, negligible vapour pressure and wide electrochemical windows,^{18,19} make the use of RTILs as solvents for amperometric gas sensors increasingly popular.^{3,20}

2 Chemicals

Ferrocene (Fe(η^5 -C₅H₅)₂, Aldrich, 98%), decamethylferrocene (Fe(η^5 -C₅(CH₃)₅)₂, DmFc, Aldrich, 98%), tetra-*n*-butylammonium perchlorate (TBAP, Fluka, Puriss electrochemical grade, 99%), acetonitrile (MeCN, Fischer Scientific, dried and distilled, 99%), agarose (Sigma, 99%), *N,N*-dimethylformamide (DMF,

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Aldrich, anhydrous 99.8%) and 1-propyl-3-methylimidazolium bistrifluoromethylsulfonylimide ([Pmim][NTf₂], kindly donated by Queen's University, Belfast) were used as received. 1,2-Diferrocenylethylene (bisferrocene, Fe(η^5 -C₅H₅)₂-(CH)₂-Fe(η^5 -C₅H₅)₂, BisFc) was synthesized using a previously reported method.¹²

3 Instrumental

All experiments were performed using a computer-controlled PGSTAT30-Autolab potentiostat (Eco-Chemie, Netherlands) and were carried out using a three-electrode arrangement consisting of a 5.0 μ m or 52.0 μ m platinum microdisc electrode as working electrodes, two silver wires as a pseudo reference and counter electrode. The platinum microdisc electrodes were polished on soft lapping pads (Kemet Ltd., UK) with alumina powders (Buehler, IL) of 1.0, 0.3 and 0.05 micron in sequence. The electrodes were immersed in distilled water and were sonicated to remove impurities. The precise radius of the micro platinum electrode was determined *via* the value of steady state current of the oxidation of ferrocene (2.0 mM) in acetonitrile (0.1 M TBAP used as supporting electrolyte), using a diffusion coefficient of 2.30×10^{-9} m² s⁻¹ at 298 K.^{21,22} Two silver wires were immersed into the hydrogel. Silver wires were employed as they provide stable voltammetric responses during the experimental time scale. Microelectrodes were adapted with small sections of disposable tips to form a cavity of volume of *ca.* 300 μ L on the electrode surface. The adapted electrodes were housed in a T-cell and studied under a controlled atmosphere.^{23,24} All experiments were performed inside a thermostatted box (previously described by Evans *et al.*)²⁵ which also functioned as a Faraday cage. The temperature was equilibrated for 0.5 hours before taking any measurements and the variation at each temperature is no more than 0.1 K. Before the electrochemical experiments, gas was run for 30 minutes to ensure equilibrium was established. O₂ and N₂ were dried by passing through dry molecular sieves (Sigma, 4 Å).

4 Preparation of DmFc and BisFc hydrogel modified micro platinum electrodes

Stock solutions were prepared by adding 2.8 mg DmFc (decamethylferrocene) and 1.2 mg BisFc (bisferrocene) in 2 mL DMF. 20 mg agarose hydrogel was boiled in 10 mL deionised water until completely dissolved. 10 μ L of the agarose and 20 μ L of the DMF stock solution were mixed and the mixture was 'drop-cast' into electrode surface. The hydrogel (*ca.* 1 mm thick) was formed *via* evaporation of solvent from the electrode surface by purging dry nitrogen for 4 hours in the T-cell at 298 K. After this, 30 μ L of 1-propyl-3-methylimidazolium bistrifluoromethylsulfonylimide ([Pmim][NTf₂]) was placed in the cavity before electrochemical measurements as detailed below.

5 Results and discussion

5.1 Characterisation of the modified electrode

In this work, the following redox reactions were analysed,



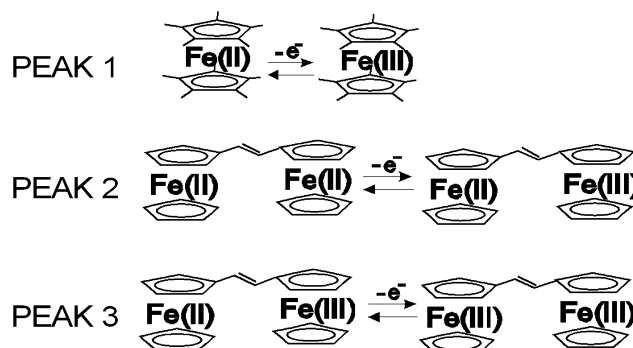
where DmFc is decamethylferrocene, BisFc is bisferrocene and each redox reaction is associated with a formal potential $E_{f,1}^0$, $E_{f,2}^0$ and $E_{f,3}^0$ respectively. See Scheme 1 for the chemical structures of the molecules studied.

The following differences in formal potential are investigated, adopting the notation given,

$$\Delta E_{12} = E_{f,2}^0 - E_{f,1}^0$$

$$\Delta E_{13} = E_{f,3}^0 - E_{f,1}^0$$

The modified electrode, the preparation of which is described in Section 3, was voltammetrically characterised in [Pmim][NTf₂] using cyclic voltammetry (CV) and square wave voltammetry (SWV). SWV is a technique that records the currents in oxidative and reductive direction and the total current is measured as a function of potential. Capacitive currents are largely reduced in SWV and hence enhanced sensitivity can be observed.²⁶ The solid line in Fig. 1a is the cyclic voltammetric response for the gel modified electrode containing DmFc and BisFc in [Pmim][NTf₂] solvent measured with a step potential of 1 mV and a scan rate of 20 mV s⁻¹ at 298 K under dry nitrogen. The concentrations of DmFc and



Scheme 1 Schematic illustration of reactions 1, 2 and 3.

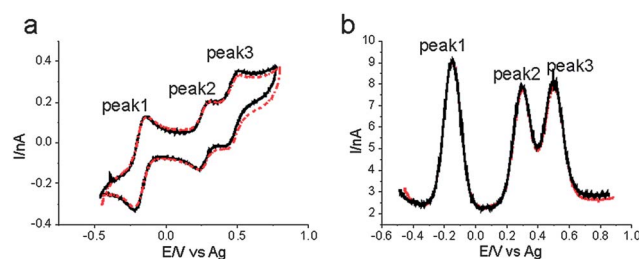


Fig. 1 (a) Cyclic voltammetry at 20 mV s⁻¹ scan rate and (b) square wave voltammetry with 200 Hz frequency, 0.1 mV step potential and 50 mV amplitude for the oxidation of decamethylferrocene and bisferrocene in PmimNTf₂ on a 50 μ m platinum electrode at 298 K over a time period of 22 hours under dry nitrogen.

BisFc in the gel were 0.258 mM and 0.20 mM respectively. No ferrocenes were added to the solvent. Three peaks, at *ca.* -0.15 V, $+0.30$ V and $+0.50$ V *vs.* Ag, were observed and labelled as peaks 1, 2 and 3 in Fig. 1a. Peak 1 is due to the oxidation of decamethylferrocene. Peaks 2 and 3 correspond to the first and second oxidation of BisFc (see Scheme 1 for molecular structures). The solid line in Fig. 1b is the SWV recorded for the same system at a frequency of 200 Hz, a step potential of 0.1 mV and an amplitude of 50 mV. In Fig. 1b peaks 1 to 3 again correspond to reactions 1 to 3.

The stability of the modified electrode was studied. It is known that stability of a detecting system is of vital importance for sensors since it influences the reproducibility and lifetimes of sensors. To investigate this, the variation of the hydrogel modified electrodes was monitored by CV and SWV for 22 hours and stored in T-cell with dry nitrogen passing through. The dashed line in Fig. 1a and b are the CV and SWV of the modified electrode after 22 hours, where only a slight drop, *ca.* 0.8%, in the peak current is observed in voltammetric responses after 22 hours, which suggests retention of DmFc and BisFc in the gel with little leaching to the ionic liquid solvent.

Further to this, the thermal response and stability were also investigated. This was performed by switching temperatures between 298 K and 308 K under dry nitrogen. This temperature range was selected because the melting point of agarose is at 309 ± 1.5 K. The temperature is controlled below this to obtain stable voltammetric responses. Fig. 2a and b show the changes of ΔE_{12} and ΔE_{13} with the temperature (see Fig. 2a for ΔE_{12} and Fig. 2b for ΔE_{13}) where the values of potential were read from square wave voltammetric responses for DmFc and BisFc. This whole process was monitored for 24 hours. Only a slight change in the peak height and peak potential in square wave voltammetry was observed. For experiments at 298 K, the differences between ΔE_{12} and ΔE_{13} are negligible. For experiments at 308 K, there are only 0.12 mV and 0.29 mV shifts in ΔE_{12} and ΔE_{13} respectively. The error bars were obtained by repeating measurements three times on the same gel. This shows that the peak separations of the system were stable with respect to change of temperature, which indicates the voltammetric responses were stable.

5.2 The temperature dependence of the voltammetry of the DmFc and BisFc hydrogel system

Fig. 3 displays square wave voltammetric responses for hydrogel confined DmFc and BisFc electrode studied in [Pmim][NTf₂] solvent over a temperature range of 298 K to 308 K, where the increase in the peak current (I_p) is primarily due to the increased diffusion coefficients (D) of the species within the hydrogel layer at higher temperatures.²⁷ The gel layer contained 0.258 mM DmFc and 0.20 mM BisFc with ferrocenes added to the ionic liquid solvent. It is noted that the shift in all formal potentials is due to the Ag pseudo reference used in this work. However the peak separations remain the same and are discussed later.

The variation of the formal potential with temperature is given by the following equation:²⁸

$$F \frac{dE_f^0}{dT} = \Delta S$$

where F is the Faraday constant, T is the temperature, E_f^0 is the formal potential and ΔS is related to the change of entropy of

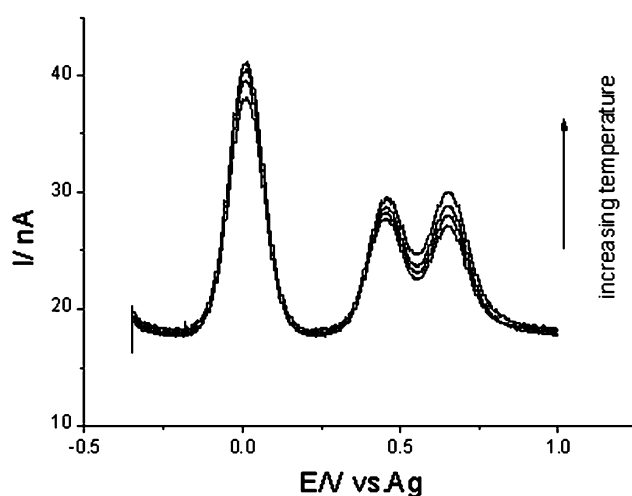


Fig. 3 Overlap of square wave voltammetry with optimized parameter (stated in the text) of oxidation of decamethylferrocene and first and second oxidation of bisferrocene over a temperature range of 298 K to 305 K under dry nitrogen.

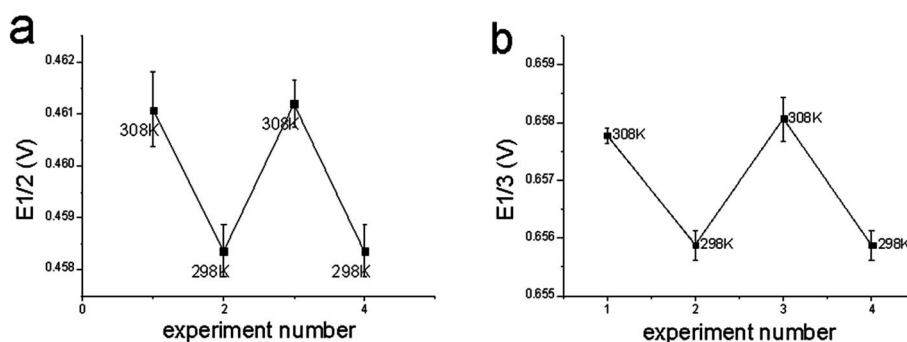


Fig. 2 Repeat measurements of ΔE_{12} (a) and ΔE_{13} (a) within a day (error bar obtained by repeating measurements three times on the same gel).

the redox reaction. If two redox couples have different temperature dependences of the formal potentials, then the potential difference of these two couples can be used to indicate temperature change.¹¹

Fig. 4 displays the measured plots of ΔE_{12} (Fig. 4a) and ΔE_{13} (Fig. 4b) against T as derived from experiments such as those shown in Fig. 3. Both ΔE_{12} and ΔE_{13} increase linearly with temperature over the range studied of 298–305 K. The relations are expressed as below,

$$\Delta E_{12} = (0.19169 \pm 0.00272) + (8.5173 \pm 0.0901) \times 10^{-4}T \quad (1)$$

$$\Delta E_{13} = (0.5719 \pm 0.0096) + (2.3201 \pm 0.3165) \times 10^{-4}T \quad (2)$$

here, T is the temperature in kelvin. The error limits of these experiments were obtained by repeating the experiment three times. These equations are used below for temperature measurement as described in the voltammetric Section 5.5.

5.3 Comparison of the ferrocene voltammetry: hydrogel confined versus ionic liquid phase ferrocene

The concentrations (c) and diffusion coefficients (D) of hydrogel confined ferrocenes and ferrocenes dissolved in the ionic liquid solution phase were compared in [Pmim][NTf₂] using chronoamperometry over a temperature range of 295 K to 308 K. The hydrogel confined ferrocenes had concentration of 0.258 mM DmFc and 0.20 mM BisFc. The solution phase

ferrocenes was studied by dissolving 2.0 mM DmFc and 0.50 mM BisFc in [Pmim][NTf₂]. Fig. 5a shows the chronoamperometric responses for (a) hydrogel confined DmFc and (b) dissolved DmFc in [Pmim][NTf₂] at 52.0 μ m Pt microdisc electrode.^{29,30} The potential was held at -310 mV vs. Ag (5 s) where no current flowed, and then stepped to a potential of -28 mV (2 s) where diffusion controlled electrolysis is fully driven. Note the data points at the first 20 ms were discarded since the currents are significantly non-diffusional.³¹ To analyse the chronoamperometric responses, Shoup and Szabo analysis was used which describes the current responses over the entire time range with an error of 0.6% (ref. 29, 32, 33 and 34) and by using this method concentration and diffusion coefficient are obtained simultaneously. This equation was imported into a home written nonlinear curve fitting function in OriginPro 8.5.1,³⁵ where the electrode radius, 52.0 μ m, was fixed and the values of concentration and diffusion coefficient were obtained *via* fitting of the experimental data. A range of c and D was input into this nonlinear function and computer performed up to 400 iterations as needed. This process automatically ceased when the resulting simulated transients reached the defined level of correlation between the experimental data points and the corresponding simulated data points. The current (I) response over the transient time at a microdisc electrode is

$$I = 4nFrDc\sqrt{t}$$

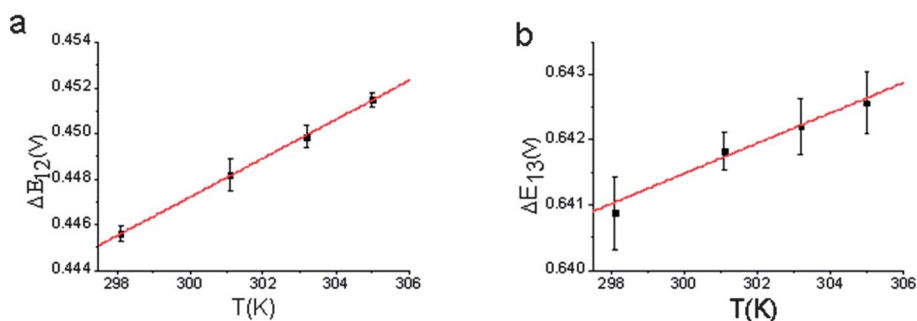


Fig. 4 Plot of (a) ΔE_{12} and (b) ΔE_{13} over a temperature range of 298 K to 305 K (experiment repeated by modifying gel three times).

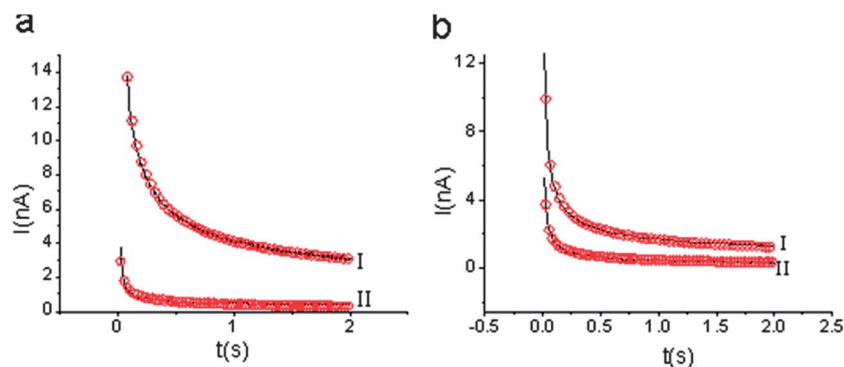


Fig. 5 Experimental (solid black) and simulated (circles) chronoamperometry for oxidation of (a) DmFc and (b) BisFc in bulk solution phase (line I) and in gel phase (line II) at a platinum electrode with radius 52.0 μ m at 298 K under dry nitrogen.

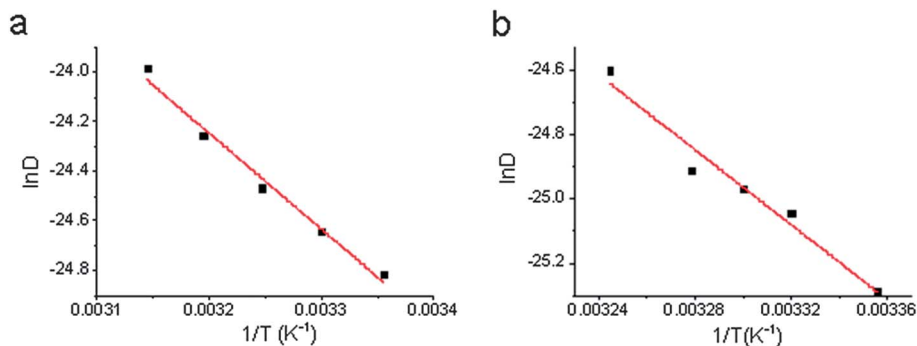


Fig. 6 Plot of $\ln D$ vs. $1/T$ for DmFc (a) dissolved and (b) hydrogel confined DmFc in gel.

where

$$f(\tau) = 0.7854 + 0.8863\tau^{-1/2} + 0.2146\exp(-0.7823\tau^{-1/2})$$

and I is the current, n is the number of electron transferred, F is the Faraday constant, c is the bulk solution concentration and $f(\tau)$ is function of τ . τ is a dimensionless time (t) parameter and is defined as $\tau = 4Dt/r_d^2$.

Fig. 5 shows typical simulated and experimental transient for hydrogel confined and ionic liquid solution phase ferrocenes in [Pmim][NTf₂]. For the hydrogel system, the concentrations of DmFc, 0.26 (± 0.01) mM, and BisFc, 0.20 (± 0.03) mM, over the temperature range studied were obtained by using Shoup and Szabo analysis. The diffusion coefficients of DmFc and BisFc from 298 K to 308 K were also determined using Shoup and Szabo analysis. The diffusion coefficients and the temperature are related *via* the Arrhenius equation:

$$D = D_\infty e^{-\frac{E_a}{RT}}$$

where D_∞ is the hypothetical value of D at an infinite temperature, T is the temperature and E_a is the activation energy for diffusion. Fig. 6a and b show the corresponding Arrhenius plots of ionic liquid [Pmim][NTf₂] solution phase and hydrogel confined DmFc, resulting in activation energies of 32.2 kJ mol⁻¹ and 43.5 kJ mol⁻¹ respectively. The same analysis was performed for the first oxidation peak of BisFc. The corresponding Arrhenius plots are shown in Fig. 7a and b, which gives values of

Table 1 Diffusion coefficient of DmFc and BisFc in [Pmim][NTf₂] at 298 K

System	DmFc ($D/\text{m}^2 \text{ s}^{-1}$)	BisFc ($D/\text{m}^2 \text{ s}^{-1}$)
Dissolved in [Pmim][NTf ₂]	1.67×10^{-11}	1.12×10^{-11}
Supported in hydrogel	4.63×10^{-12}	4.51×10^{-12}

activation energy of 48.7 kJ mol⁻¹ and 75.8 kJ mol⁻¹ for ionic liquid solution phase BisFc and hydrogel confined BisFc in [Pmim][NTf₂] respectively. The values at 298 K are summarised in Table 1, where the diffusion activation energy for DmFc is lower than that of BisFc in the same environment which is likely due to the smaller molecular size of DmFc.¹⁸ That molecules in hydrogel have a higher diffusion activation energy reflects the fact that molecules find it much more difficult to move through the pores of the gel as compared to fluid solutions and is fully consistent with the general immobilisation approach advocated.

5.4 Investigation of the hydrogel modified electrode in dry oxygen

The hydrogel modified electrode was next incorporated into an oxygen sensing system. Experiments were performed by purging dry oxygen into the system containing an electrode with a layer of hydrogel containing confined ferrocenes below a layer of pure [Pmim][NTf₂] of *ca.* 2.2 mm was used as solvent. Fig. 8a and b shows the cyclic voltammetric and square wave voltammetric responses respectively for the hydrogel confined ferrocenes

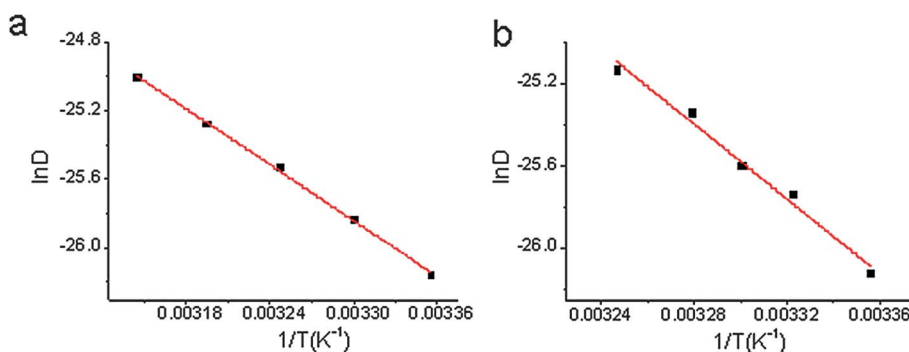


Fig. 7 Plot of $\ln D$ vs. $1/T$ of (a) dissolved and (b) hydrogel confined BisFc.

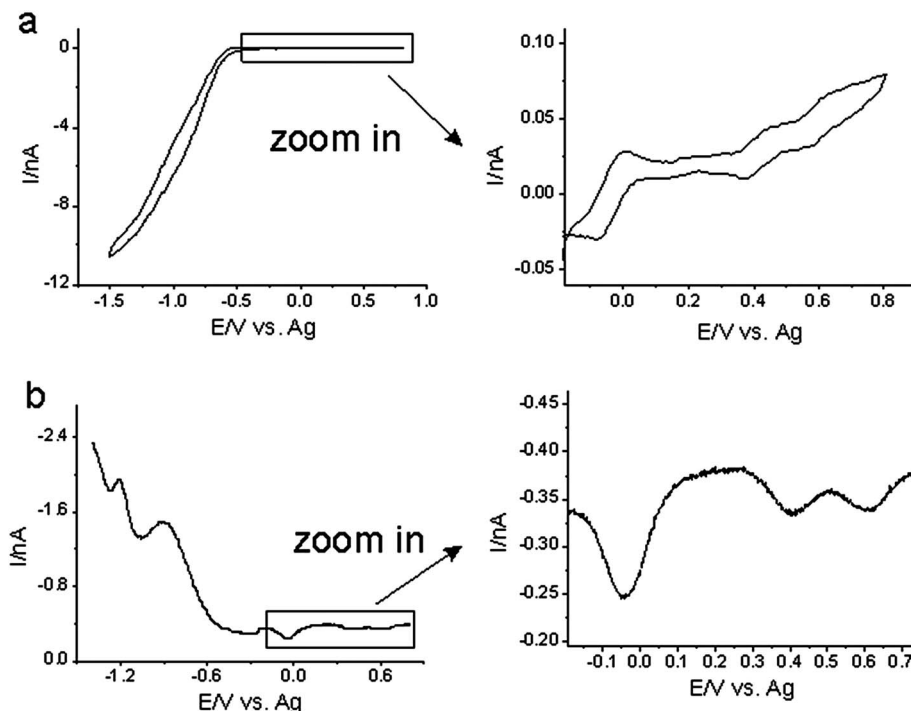
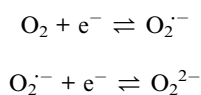


Fig. 8 (a) Cyclic voltammetry at scan rate of 20 mV s^{-1} and (b) square wave voltammetry of 50 Hz frequency, 1 mV step potential and 50 mV amplitude of reduction of oxygen and oxidation of decamethylferrocene and BisFc at 298 K on hydrogel modified platinum electrode with radius $5.0 \mu\text{m}$ under dry oxygen.

(DmFc and BisFc) in the presence of oxygen. The peaks due to DmFc and BisFc are enlarged for clarity. The peaks observed at -1.07 V and -1.27 V are assigned to be the first and second reduction of oxygen respectively:^{36,37}



However, in presence of proton impurities, the nucleophilic ion O_2^{2-} generated *via* reduction can react leading to irreversible reaction.

In order to get a voltammetric response, oxygen has to diffuse through the RTIL and then through the hydrogel. Thus

Table 2 Diffusion coefficient and concentration of oxygen in the hydrogel

Temperature (K)	Diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)	Concentration (mM)
298	9.18×10^{-10}	3.65
301	1.08×10^{-9}	3.87
303	1.16×10^{-9}	3.80
305	1.27×10^{-9}	3.80

the hydrogel is oxygen permeable and well-defined voltammetry of oxygen is seen at the modified electrode. Diffusion coefficient and concentration data was obtained from potential step chronoamperometric experiments. The potential-step chronoamperometry was undertaken using a sample time of 0.005 s.

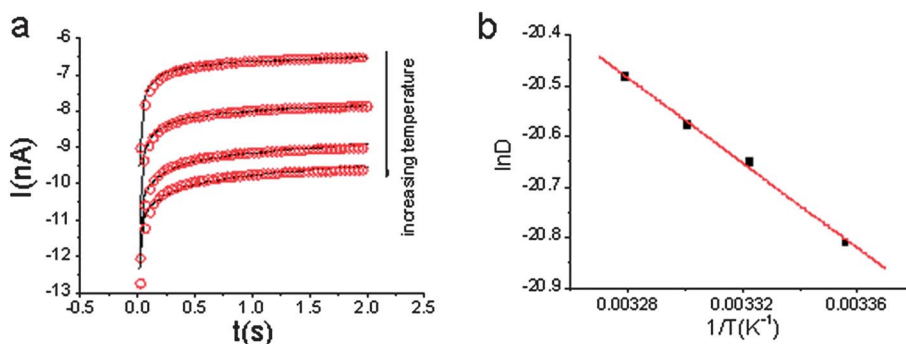


Fig. 9 (a) Experimental (solid) and simulated (circles) chronoamperometry for the reduction of oxygen in PminNTf₂ over a temperature range of 298 K to 303 K under dry oxygen with hydrogel modified electrode with $5.0 \mu\text{m}$ radius. (b) Plot of $\ln D$ vs. $1/T$ for oxygen where T is read from the thermostat.

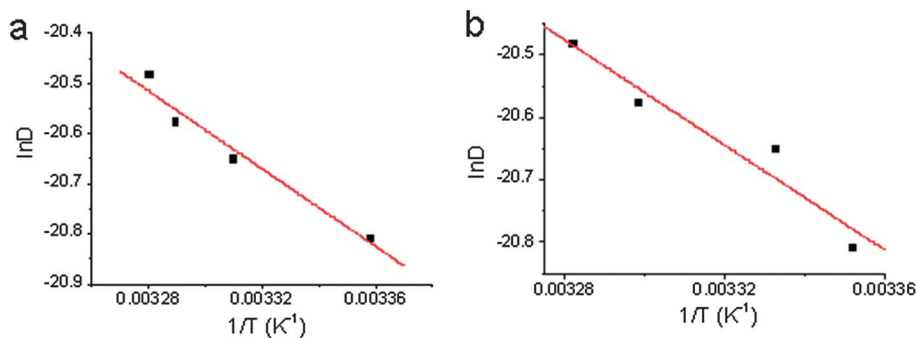


Fig. 10 (a) Plot of $\ln D$ vs. $1/T$ for oxygen where T is calculated from eqn (1). (b) Plot of $\ln D$ vs. $1/T$ for oxygen where T is calculated from eqn (2).

The system was pretreated by holding the potential at -280 mV where no faradaic current was passed for 5 s, after which the experimental transient (Fig. 9a) was obtained by stepping to a potential of -1270 mV where the reduction of oxygen was fully driven for 2 s.²⁹ The diffusion coefficient and concentration of oxygen in the hydrogel were simultaneously analysed as a function of temperature by fitting the Shoup and Szabo equation to chronoamperometry. This was carried out over the temperature range of 297 K to 305 K. Table 2 shows the diffusion coefficient and concentration at different temperatures.

The concentration of oxygen (shown in Table 2) in hydrogel was found to be $3.80 (\pm 0.15)$ mM and insensitive to temperatures. The essential independence of temperature was in good agreement with those published in other ionic liquid systems²¹ and in aqueous systems.³⁸ This fortuitous observation may be valuable in simplifying oxygen sensing investigations where the only temperature sensitive variable is the diffusion coefficient.

To investigate the temperature dependence of the diffusion coefficient, Arrhenius plots were used. The natural logarithm of the diffusion coefficient of oxygen was plotted against reciprocal temperature as shown in Fig. 9b, which gave a good linear trend and activation energy of 34.8 kJ mol^{-1} . As for comparisons to the solution phase ionic liquids with unmodified electrode, the activation energies for oxygen were also compared to literature values, which lie within a range of $21.5\text{--}27.9 \text{ kJ mol}^{-1}$ for ionic liquids with similar viscosity.^{11,39} The solution phase system has a smaller activation energy than the hydrogel confined system, which again suggests that the structure of the gel somewhat impedes the diffusion of oxygen related to the ionic liquid.

5.5 Test of the *in situ* voltammetric thermometer

The hydrogel confined DmFc and BisFc electrode was studied in $[\text{Pmim}][\text{NTf}_2]$ over a temperature range of 299 K to 305 K. The gel layer contained 0.258 mM DmFc and 0.2 mM BisFc with no ferrocenes added to the ionic liquid solvent. The whole system described above was tested under dry oxygen (99.9%). Square wave voltammetry was measured at increasing temperatures and the potential differences between peaks 1 and 2, and peaks 1 and 3 were measured. A similar relationship between peak potential differences and T was observed as in eqn (1) and (2).

By using these equations, T may also be obtained either by substituting ΔE_{12} into eqn (1) and ΔE_{13} into eqn (2) or

alternatively directly read from external thermometer. In the former way an Arrhenius plot of oxygen can be studied using peak potential differences so giving a purely voltammetric measurement of the diffusion coefficient. Fig. 10a and b show the Arrhenius plots for oxygen where T is calculated by substituting ΔE_{12} into eqn (1) and ΔE_{13} into eqn (2) respectively, where the measured activation energy are $32.2 \pm 2.3 \text{ kJ mol}^{-1}$ and $34.9 \pm 3.2 \text{ kJ mol}^{-1}$ from each approach. These values are highly comparable to the activation energy of $34.8 \pm 1.6 \text{ kJ mol}^{-1}$ (see above) achieved from external readings. Table 3 compares activation energy for oxygen from different approaches. It is seen that the values are identical within experimental error which indicates the modified electrodes can be realistically used as a temperature probes for gas sensing.

Table 3 Activation energy for diffusion of oxygen in BisFc and DmFc modified hydrogel system (temperature read from the thermostat and calculate from eqn (1) and (2))

	Activation energy of oxygen/ kJ mol^{-1}
T read from thermostat	34.8
T calculated from eqn (1)	32.2
T calculated from eqn (2)	34.9

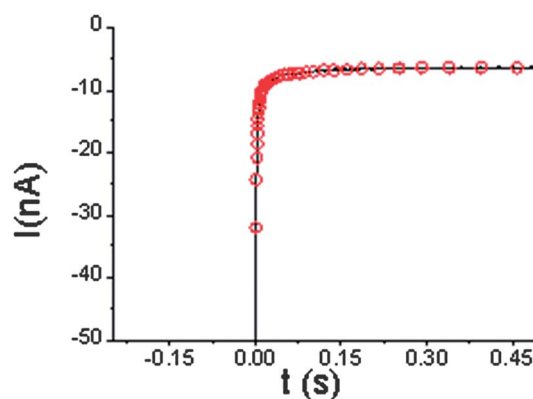


Fig. 11 Experimental (solid) and simulated (circles) chronoamperometry for the reduction of oxygen at μPt electrode in PminNTf_2 over a temperature range of 298 K to 303 K under dry oxygen.

The oxygen concentration detected from the gel modified electrode was 3.8 mM and again no significant variation was found over the temperature range studied. For comparison, an experiment was carried out in pure [Pmim][NTf₂] with dry oxygen passing through at 298 K and Shoup and Szabo analysis was used to determine the oxygen concentration (see Fig. 11), which was *ca.* 5.2 mM. This is also compared to saturated oxygen concentrations dissolved in various ionic liquid, which show a range of 4.4 to 5.2 mM.^{11,40} The significantly reduced solubility in the hydrogel reflects the complex network.^{41,42}

6 Conclusions

A method of modifying DmFc and BisFc onto the electrode surface using agarose hydrogel has been demonstrated and the stability of the recently modified electrode has been established. The immobilised DmFc and BisFc were studied *via* CV and SWV where 3 peaks were observed. Two temperature sensitive peak to peak separations (ΔE_{12} and ΔE_{13}), were used as a temperature sensor over a temperature range of 298 K to 308 K. The temperature coefficients of ΔE_{12} and ΔE_{13} were found to be 0.84 mV K⁻¹ and 0.52 mV K⁻¹ respectively. Activation energies for diffusion of DmFc and BisFc in the hydrogel modified electrode were analysed *via* the Arrhenius equation. This DmFc and BisFc hydrogel modified electrode was used in oxygen analysis. The concentration of oxygen in the hydrogel, 3.8 mM, was found to be insensitive to the temperature change when studied in [Pmim][NTf₂]. The activation energy of oxygen also was studied by plotting log diffusion coefficient as a function of reciprocal of temperature. Good agreement in activation energy for diffusion of oxygen was found between the results obtained with temperature measured from an external thermometer, 34.8 kJ mol⁻¹, and those calculated from voltammetric measurements, 32.2 kJ mol⁻¹ from ΔE_{12} and 34.9 kJ mol⁻¹ from ΔE_{13} . The results clearly indicate this system can be realistically used as a temperature indicator for amperometric oxygen sensors and other gas sensors overcoming the ferrocene volatility problem and allowing the ready interpretation of current in terms of concentrations.

Acknowledgements

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